ACCSYS

Field Operations Platform

Algorithm & Specification Reference

Form Algorithms · Gate Decision Logic · Notification Matrix
Table Dependencies · Inter-Relation Map · Bugs & Missing

Ratal Group · September 2026

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Reviewed by

Comments / Corrections on this document:

SECTION 1 — Form Algorithms

FormSubConClaim — SubCon Payment Request

[QR Auth] [App Only] [3 Gates Active]

Actor: Sub-Contractor
Route: /forms/subcon-claim
Writes To: subcon_claims
Reads From: outgoing_purchase_orders, sites, contractors, qr_tokens

ALGORITHM

```
TRIGGER:  SubCon opens App → taps "Request Payment"

STEP 1 — QR Authentication [G5]
  scan_qr() → resolve qr_tokens.contractor_id, entity_id
  IF token expired OR inactive → BLOCK "Invalid QR. Contact DH."

STEP 2 — PO Existence Check [G2]
```

```
  SELECT * FROM outgoing_purchase_orders
  WHERE contractor_id=? AND status="OPEN" AND balance > 0
  IF result=[] → BLOCK "No open PO. Contact your DH."

STEP 3 — PO Acknowledgement Check [G1]
  IF outgoing_purchase_orders.acknowledged = FALSE → BLOCK
    show_po_details() → SubCon taps "Acknowledge"
    UPDATE outgoing_purchase_orders SET acknowledged=TRUE
    THEN allow payment request

STEP 4 — PO Selection (FIFO)
  Auto-select: MIN(po_date) from open POs  ← oldest first
  DH override: select by Site# in Money Request

STEP 5 — Form Fields (3 taps only)
  ① Site (auto-filled from PO)
  ② Amount (SAR) — validate: amount ≤ po.balance
  ③ Photo attachment (evidence of work done)
  IF amount > po.balance → WARN "Exceeds PO balance (SAR X remaining)"

STEP 6 — Submit
  INSERT INTO subcon_claims {contractor_id, site_id, po_id,
    amount, photo_url, status:"PENDING", claim_number}
  NOTIFY DH: In-App + Telegram "New payment request — SAR [amount]"
```

INTER-LINKS

- → REQUIRES → outgoing_purchase_orders (open PO with balance)
- → REQUIRES → PO Acknowledged (G1)
- → TRIGGERS → DH reviews → Payment Certificate creation
- → BEFORE → FormMoneyRequest (SubCon line)
- → READS → FormSiteCompletion (basis for DH certification)

TEMPLATE

None at submission. SubCon Payment Certificate generated after DH certifies (⚠ needs rebuild).

CODE ISSUES

⚠ CODE PATCH NEEDED: Simplify to 3 taps. Remove claim_type, milestones, hours, hourly_rate — too complex for field SubCons. DH fills these in Payment Certificate.

⚙ FormMoneyRequest — Main Payment Engine

[QR Auth] [App + Excel UserForm] [2 Known Bugs]

Actor: DH / PM / Supervisor (App OR Excel UserForm)

Route: /money-requests/new

Writes To: money_requests, money_request_lines

Reads From: employees, contractors, suppliers, sites, projects, departments, outgoing_purchase_orders, payment_certificates, subcon_claims, bank_accounts

ALGORITHM

```
TRIGGER:  DH opens App / Excel UserForm → New Money Request

STEP 1 — QR Auth [G5]
  App: scan_qr() → dh_id, entity_id
  Excel: VBA Login → Supabase session token

STEP 2 — Bank Account Routing (Auto)
  IF department = "TISU"          → default bank = ANB-77
  IF department = NISU/CISU/ITSU  → default bank = ANB-39
  ELSE                            → select ANB-15, ANB-18, ANB-11
  DH can override per line

STEP 3 — Line Entry (multi-line)
  payment_type: [Employee | Contractor | Supplier | Vendor |
                 Sub-Contractor | Local Supplier | Petty Cash]

  IF payment_type = "Sub-Contractor":
    REQUIRE Site# → load open POs for site + subcon
    CHECK payment_certificates WHERE status="CERTIFIED" [G3]
    IF no certificate → BLOCK "Create Payment Certificate first"
```

```
    IF amount > cert.certified_amount → BLOCK "Exceeds certified amount"

  IF payment_type = "Petty Cash":
    auto_route_bank(department) → ANB-77 or ANB-39

STEP 4 — Submit
  INSERT INTO money_requests {header fields}
  INSERT INTO money_request_lines {line_type, supplier_name,
    account_code, iban, amount}
  ⚠ BUG: project_id and site_id NOT saved to money_request_lines

STEP 5 — Notifications
  EMAIL Accounts: "New Money Request #MR-XXXX — SAR [total]"
  TELEGRAM: instant alert to Accounts channel
  IN-APP Accounts: MR appears in payment queue

EXCEL USERFORM PARALLEL PATH:
  VBA: load_projects() ← from Supabase projects table
  VBA: load_sites(project_id) ← from Supabase sites table
  VBA: load_subcons(site_id) ← from Supabase contractors table
  DH fills amounts → VBA: submit_to_supabase(money_requests)
  ← Same table as App. Accounts sees one queue regardless of interface.
```

INTER-LINKS

- → REQUIRES (SubCon lines) → payment_certificates (G3)
- → REQUIRES (SubCon lines) → outgoing_purchase_orders (open)
- → TRIGGERS → Accounts payment queue
- → UPDATES → outgoing_purchase_orders.balance (after payment)
- → PARALLEL → Excel UserForm (identical Supabase backend)

TEMPLATE

Money Request Template □ MISSING — printable MR for Accounts physical records.

CODE ISSUES

⚠ BUG 1: project_id in EMPTY_LINE state NOT saved to money_request_lines. Fix: add project_id: l.project_id to lineRows map in FormMoneyRequest.jsx.

⚠ BUG 2: No Site# field on money_request_lines. Without Site#, cannot auto-resolve which open PO to apply SubCon payment against.

□ Payment Certificate — SubCon Work Progress Certification

[App + Desktop] [Template Rebuild Needed ⚠]

Actor: DH (certifies) → Accounts (reads)
Route: DH Dashboard / Payment Certification screen
Writes To: payment_certificates
Reads From: subcon_claims, outgoing_purchase_orders, site_completion_records

ALGORITHM

```
TRIGGER:  DH reviews SubCon payment request → opens Payment Certificate

STEP 1 — Load Context
  load_subcon_claim(claim_id): requested amount, site, photo
  load_po_status(po_id):
    po_value   = outgoing_purchase_orders.value
    prev_paid  = SUM(money_request_lines.amount WHERE po_id=?)
    po_balance = po_value − prev_paid
  load_site_completion(site_id): milestone completion %

STEP 2 — DH Certifies
  DH enters: work_progress_pct (0–100%)
  System calculates:
    earned_amount      = po_value × work_progress_pct
    previously_paid    = prev_paid (from DB)
    certifiable_amount = earned_amount − previously_paid
    this_payment       = MIN(claim.amount, certifiable_amount)

  IF this_payment ≤ 0 → WARN "Progress % does not support further payment"
  DH can override with manual certified_amount (with audit note)
```

```
STEP 3 — Save & Generate
  INSERT INTO payment_certificates:
    {subcon_claim_id, po_id, work_progress_pct, po_value,
     prev_paid, certified_amount, status:"CERTIFIED",
     certified_by:dh_id, certified_at:NOW()}
  GENERATE: SubCon Payment Certificate DOCX ⚠ (rebuild needed)
  NOTIFY Accounts: Certificate ready → MR can now be submitted

STEP 4 — Unlocks Money Request [G3 satisfied]
  payment_certificates.status = "CERTIFIED"
  DH can now submit FormMoneyRequest for this SubCon
```

INTER-LINKS

- → REQUIRES → subcon_claims (SubCon payment request must exist)
- → READS → site_completion_records (work progress basis)
- → UNLOCKS → FormMoneyRequest SubCon line (G3)
- → GENERATES → SubCon Payment Certificate DOCX ⚠

TEMPLATE

SubCon Payment Certificate DOCX ⚠ — Needs rebuild: add work_progress_pct, PO status block (Total PO / Previously Paid / This Payment / Balance After), earned vs certified amount.

□ FormFieldPayment — Cash Advance to Employee

[App] [Auto JE: Dr 1310 / Cr Petty Cash]

Actor: DH / PM
Route: /forms/field-payment
Writes To: employee_advances (txn_type: ADVANCE_IN), journal_entries
Reads From: employees, bank_accounts, sites

ALGORITHM

```
TRIGGER:  DH issues cash to employee on-site

STEP 1 — Select Recipient
  Currently: Employee only
  ⚠ NEEDS EXTENSION: SubCon payee + PO linkage (code patch session)
  Select: employee_id, amount (SAR), site_id, bank_account, notes

STEP 2 — Post Journal Entry (Auto)
  post_journal_entry():
    Dr: 1310 Employee Advance   [amount]
    Cr: ANB-77 / ANB-39 Petty Cash   [amount]
    ref: advance_id, employee_id, site_id, date

STEP 3 — Save Record
  INSERT INTO employee_advances:
    {employee_id, amount, site_id, txn_type:"ADVANCE_IN",
     confirmed:FALSE, issued_by:dh_id, issued_at:NOW()}

STEP 4 — Trigger Payment Validation [G6]
  NOTIFY Employee: In-App "SAR X advance issued. Confirm receipt."
  Employee must complete FormPaymentValidation
  Until confirmed: flagged on DH dashboard as "Unconfirmed Advance"
```

INTER-LINKS

- → TRIGGERS → FormPaymentValidation (employee confirms receipt)
- → CLEARS AGAINST → FormExpenseClaim (month-end reconcile)
- → POSTS → Auto JE Dr 1310 / Cr Petty Cash

TEMPLATE

None.

CODE ISSUES

⚠ PATCH 3 NEEDED: Extend to SubCon as payee type + link to open Outgoing PO.

FormPaymentValidation — Employee Confirms Advance Receipt

[QR Auth] [App]

Actor: Employee (recipient)
Route: /forms/payment-validation
Writes To: employee_advances (UPDATE confirmed=TRUE)
Reads From: employee_advances (unconfirmed records)

ALGORITHM

```
TRIGGER:  Employee receives In-App notification about advance

STEP 1 — Load Pending Confirmation
  SELECT * FROM employee_advances
  WHERE employee_id=? AND confirmed=FALSE AND txn_type="ADVANCE_IN"
  IF none → show "No pending confirmations"

STEP 2 — Employee Confirms
  Shows: advance amount, issued by, date
  Employee: uploads photo of signed payment voucher
  Employee: taps Confirm

STEP 3 — Update Record
  UPDATE employee_advances SET:
    confirmed=TRUE, confirmation_photo=[url],
    confirmed_at=NOW(), confirmed_by=employee_id

STEP 4 — 24h Monitoring
  IF not confirmed within 24h:
    ALERT DH: "[Employee] has not confirmed SAR X advance"
    Flag persists on dashboard until confirmed
```

INTER-LINKS

- → REQUIRES → FormFieldPayment (advance record must exist)
- → CLEARS FLAG ON → DH Dashboard
- → STORES → document_vault (signed voucher photo)

TEMPLATE

None.

FormFoodAllowance — Daily Food Allowance

[QR Auth] [App] [SAR 30/day/person default]

Actor: Employee / Supervisor
Route: /forms/food-allowance
Writes To: food_allowances
Reads From: employees, sites, departments

ALGORITHM

```
TRIGGER:  Employee / supervisor submits food allowance for site team

STEP 1 — QR Auth [G5]

STEP 2 — Form Fields
  no_persons   (number of people eating)
  daily_rate   (default SAR 30 — editable if site rate differs)
  no_days      (duration on remote site)
  site_id      (auto from QR or picker)
  notes

STEP 3 — Calculate
  total_amount = no_persons × daily_rate × no_days

STEP 4 — Submit & DH Approval
  INSERT INTO food_allowances {… status:"PENDING"}
  NOTIFY DH: In-App "Food allowance pending — SAR [total]"
  DH: Approve → UPDATE status="APPROVED" → feeds Payroll
  DH: Reject  → NOTIFY Employee with reason
```

INTER-LINKS

- → REQUIRES APPROVAL → DH Dashboard (Gate)
- → FEEDS → Payroll processing (monthly)
- → COMPILED INTO → FormExpenseClaim (approved amounts)

TEMPLATE

None.

FormOvertime — Overtime Request

[QR Auth] [App] [NORMAL/FRIDAY/HOLIDAY=1.5× | TRAVEL=1.0×]

Actor: Employee / Supervisor
Route: /forms/overtime
Writes To: overtime_records
Reads From: employees, sites, departments

ALGORITHM

```
TRIGGER:  Employee / supervisor submits OT record

STEP 1 — QR Auth [G5]

STEP 2 — Form Fields
  employee_id, site_id, date, hours_worked
  ot_type: [NORMAL | FRIDAY | HOLIDAY | TRAVEL]

STEP 3 — Calculate
  multiplier = CASE ot_type
    WHEN TRAVEL → 1.0
    ELSE        → 1.5
  ot_pay = employee.hourly_rate × multiplier × hours_worked

STEP 4 — Submit
  INSERT INTO overtime_records {… status:"PENDING"}
  NOTIFY DH: In-App "OT pending: [employee] [hours]h [type]"

STEP 5 — DH Approval Gate [G7]
  IF NOT approved → excluded from payroll
  IF approved → UPDATE status="APPROVED" → included in payroll run
```

INTER-LINKS

- → GATE → DH must approve before payroll inclusion (G7)
- → FEEDS → Payroll (monthly run)
- → CROSS-CHECK POSSIBLE → FormSiteCompletion (flag OT on inactive sites)

TEMPLATE

None.

FormDailyExpense — Daily Field Expense Entry

[QR Auth] [App] [Accumulates to month-end claim]

Actor: Employee
Route: /forms/daily-expense
Writes To: daily_expenses
Reads From: employees, sites, expense_categories

ALGORITHM

```
TRIGGER:  Employee submits daily expenses from field

STEP 1 — QR Auth [G5] → resolve employee_id, site_id

STEP 2 — Form Fields
  date, expense_category, amount, receipt_photo, notes

STEP 3 — Submit
  INSERT INTO daily_expenses {employee_id, site_id, date,
    category, amount, photo_url, claimed:FALSE}
```

```
ACCUMULATION LOGIC (month-end):
  FormExpenseClaim auto-pulls:
    SELECT SUM(amount) FROM daily_expenses
    WHERE employee_id=? AND MONTH(date)=claim_month
    AND claimed=FALSE
  Groups by category → populates expense_claim_lines[]
  Marks daily_expenses.claimed=TRUE after claim submitted
```

INTER-LINKS

- → ACCUMULATES TO → FormExpenseClaim (monthly compilation)
- → STORES → document_vault (receipt photos)

TEMPLATE

None.

□ FormExpenseClaim — Monthly Expense Claim

[QR Auth] [Telegram] [Email] [Auto JE] [Template Built □]

Actor: Employee → DH Approves → Accounts Pays
Route: /forms/expense-claim
Writes To: expense_claims, expense_claim_lines, journal_entries
Reads From: daily_expenses, employees, sites, food_allowances, employee_advances

ALGORITHM

```
TRIGGER:  Month-end OR employee initiates claim

STEP 1 — Cut-off Check
  IF today > monthly_cutoff_date → "Claim closed. Wait for next cycle."

STEP 2 — Auto-Compile Lines
  Pull all daily_expenses WHERE employee_id=? AND claimed=FALSE
  Group by category → expense_claim_lines[]
  Include: food_allowances (approved), outstanding advances

STEP 3 — Submit
  INSERT INTO expense_claims {employee_id, period, total, status:"SUBMITTED"}
  INSERT INTO expense_claim_lines {claim_id, category, amount, …}
  UPDATE daily_expenses SET claimed=TRUE WHERE included
  NOTIFY DH: Telegram + In-App "Expense claim — SAR [total]"

STEP 4 — DH Approval [G8]
  Approve → status="APPROVED"
    → post_journal_entry(): Dr Expense / Cr Employee Payable
    → GENERATE: Expense Claim DOCX ▫
    → NOTIFY Employee: Email + In-App "Approved"
    → Accounts payment queue updated
  Reject → status="REJECTED" → Employee corrects → resubmit
```

INTER-LINKS

- → COMPILES FROM → FormDailyExpense (all unclaimed entries)
- → INCLUDES → FormFoodAllowance (approved amounts)
- → RECONCILES → FormFieldPayment (cash advances at month-end)
- → GENERATES → Expense Claim DOCX □
- → GATE → DH approval (G8) before Accounts can see/pay

TEMPLATE

Expense Claim DOCX □ — Built. Employee, period, lines, totals, DH approval block.

□ FormCarMaintenance — Vehicle Log

[QR Auth] [App] [FUEL · MAINTENANCE · INCIDENT · INSPECTION]

Actor: Driver / Employee
Route: /forms/car-maintenance
Writes To: vehicle_maintenance

Reads From: vehicles, employees, sites

ALGORITHM

```
TRIGGER:  Driver / employee records vehicle event

STEP 1 — QR Auth [G5]

STEP 2 — Form Fields
  vehicle_id (from vehicles master)
  record_type: [FUEL | MAINTENANCE | INCIDENT | INSPECTION]
  date, mileage, cost, notes, photo_attachment
  IF INCIDENT: damage description, third party, police report
  IF INSPECTION: next_inspection_date, pass/fail result

STEP 3 — Submit
  INSERT INTO vehicle_maintenance {vehicle_id, record_type,
    date, mileage, cost, notes, photo_url}

REPORTING → Vehicle Expense Report, Incident Log, Inspection Tracker
```

INTER-LINKS

- → READS → vehicles (master table)
- → FEEDS → Vehicle Expense Report (cost per vehicle/period)
- → INCIDENT → document_vault (police report, photos)

TEMPLATE

None.

FormSiteCompletion — Site Progress Recording

[QR Auth] [App] [Key: basis for SubCon payment certification]

Actor: PM / Supervisor
Route: /forms/site-completion
Writes To: site_completion_records, milestone_records
Reads From: sites, milestones, projects, subcon_assignments

ALGORITHM

```
TRIGGER:  PM / Supervisor records site progress (daily or milestone)

STEP 1 — QR Auth [G5] → resolve site_id, project_id

STEP 2 — Form Fields
  site_id, project_id, date
  overall_progress_pct (0–100%)
  milestone_records[]:
    {milestone_id, status: COMPLETE/IN_PROGRESS/NOT_STARTED,
     completion_pct, notes, photo}

STEP 3 — Submit
  INSERT INTO site_completion_records {site_id, project_id,
    date, overall_pct, submitted_by}
  UPSERT INTO milestone_records {milestone_id, status, pct}

DOWNSTREAM USE:
  DH opens Payment Certificate → load_site_completion(site_id)
  System shows current overall_pct → DH uses as certification basis
  CRITICAL: site completion % justifies certified SubCon payment.
  DH should not certify more than % warrants.
```

INTER-LINKS

- → FEEDS → Payment Certificate (DH certification basis)
- → FEEDS → DH Dashboard (site progress view)
- → FEEDS → Site Progress Report (Management)
- → CROSS-CHECK POSSIBLE → FormOvertime (flag OT on inactive sites)

TEMPLATE

None.

□ FormPORequest — Purchase Order Request

[App + Desktop] [Email] [Outgoing PO Template □]

Actor: DH / PM
Route: /forms/po-request
Writes To: po_requests → outgoing_purchase_orders (after Accounts approval)
Reads From: departments, employees, projects, sites, contractors, suppliers, items

ALGORITHM

```
TRIGGER:  DH needs to engage SubCon / Supplier for work or materials

STEP 1 — Cascade Selection
  Department → Employee → Project → Site → PO Type
  → Supplier/Contractor → Line Items

STEP 2 — Line Items
  For each item: description, qty, unit, unit_price → line_total
  Header total = SUM(line_totals) + VAT (15%)

STEP 3 — Submit PO Request
  INSERT INTO po_requests {dept, project, site, contractor_id,
    items[], total, status:"PENDING"}
  NOTIFY Accounts: Email + In-App "New PO Request — SAR [total]"

STEP 4 — Accounts Approves
  INSERT INTO outgoing_purchase_orders:
    {contractor_id, site_id, value:total, balance:total,
     status:"OPEN", po_number, po_date, acknowledged:FALSE}
  GENERATE: Outgoing PO DOCX □ → PDF → sent to SubCon
  NOTIFY SubCon: In-App + Telegram + Email
    "New PO #X issued. Tap to acknowledge."
  ← Triggers G1 (PO acknowledgement gate)
```

INTER-LINKS

- → GENERATES → outgoing_purchase_orders record (open, balance=po_value)
- → TRIGGERS → G1 (SubCon must acknowledge before payment request)
- → GENERATES → Outgoing PO Template □
- → UNLOCKS → SubCon Payment Request (G2 can now pass)

TEMPLATE

Outgoing PO DOCX □ — Built. Sent to SubCon with PO details on issuance.

✈ FormPayment — Travel & Field Distribution (Shareable)

[QR Auth] [Shareable Link] [Entity-aware (RAT/GWT vs ACCSYS)]

Actor: DH / PM / Authorized party via shared link
Route: /forms/payment
Writes To: payments (travel or field distribution)
Reads From: employees, contractors, bank_accounts, qr_tokens

ALGORITHM

```
TRIGGER:  Shared link sent OR DH opens directly

STEP 1 — Entity Detection
  detect_entity(qr_token / url_param) → entityCode
  IF RAT or GWT → show Travel Payment fields
  IF ACCSYS     → show Field Distribution fields

ENTITY = RAT/GWT — Travel Payment
  Fields: travel_type [per diem|hotel|flight|transport|visa|misc]
  traveler_id, destination, travel_dates, receipt_photo
  bank_account selection, amount (SAR)

ENTITY = ACCSYS — Field Distribution
  Fields: payee_type [employee|sub-contractor|supplier]
  payee_id, site_id, bank_account, amount, photo, notes
```

```
STEP 2 — Submit
  INSERT INTO payments {entity_code, payment_type,
    payee_id, amount, bank_account, …}
  Auto-route to appropriate ledger account by payment_type
```

INTER-LINKS

- → ENTITY-AWARE → different fields for RAT/GWT vs ACCSYS
- → FEEDS → Ledger accounts (by payment_type)
- → BANK ACCOUNTS → ANB-77, ANB-39, ANB-15, ANB-18, ANB-11, OTHER

TEMPLATE

None.

SECTION 2 — Gate Decision Logic

All gates listed below are hard system blocks. Each shows: condition checked, what is blocked, resolution path, and override rules.

G1 — PO Acknowledgement Gate

Purpose: Ensures SubCon is aware of PO terms before requesting payment. Prevents "I did not know" disputes.
Override: None. DH cannot bypass on SubCon behalf.

DECISION LOGIC

```
WHEN:     SubCon submits payment request
CHECK:
  ack = outgoing_purchase_orders.acknowledged
        WHERE contractor_id = X

  IF ack = FALSE:
    BLOCK payment request
    ACTION: show PO details → SubCon taps Acknowledge
    UPDATE outgoing_purchase_orders SET acknowledged=TRUE
    THEN: payment request unlocked → proceed to G2

  ELSE: proceed directly to G2
```

G2 — PO Balance & Open Status Gate

Purpose: Prevents payment requests when no open PO exists or balance is zero. FIFO selects oldest PO; DH can override by Site#.
Override: DH can select a different PO by choosing Site# in Money Request. FIFO is a suggestion, not absolute.

DECISION LOGIC

```
WHEN:     SubCon submits payment request (after G1 passes)
CHECK:
  open_pos = SELECT * FROM outgoing_purchase_orders
             WHERE contractor_id=X AND status="OPEN" AND balance > 0

  IF open_pos = []:
    BLOCK "No open PO with balance. Contact DH."
    EXIT

  FIFO SELECTION:
    selected_po = MIN(open_pos, key=po_date)  ← oldest first
    available   = selected_po.value − SUM(payments WHERE po_id=?)

  AMOUNT VALIDATION:
    IF requested_amount > available:
      WARN "Amount (SAR X) exceeds PO balance (SAR Y). Reduce."

  EXHAUSTION CHECK → triggers G4:
    IF available = 0 AND invoice_not_attached:
      BLOCK → redirect to Invoice Submission gate (G4)
```

G3 — Payment Certificate Required (Before SubCon Money Request)

Purpose: DH must witness site progress before authorizing SubCon payment. Prevents paying for work not yet done. Certified amount = PO_value × work_progress_pct − previously_paid.

Override: DH can manually override with justified note. All overrides are audit-logged.

DECISION LOGIC

```
WHEN:     DH adds Sub-Contractor line to FormMoneyRequest
CHECK:
  cert = SELECT * FROM payment_certificates
         WHERE contractor_id=X AND site_id=Y
         AND status="CERTIFIED"
         AND certified_at >= subcon_claim.submitted_at

  IF cert = NULL:
    BLOCK "Create Payment Certificate first."
    SHOW LINK → Payment Certificate screen
    EXIT

  AMOUNT CAP:
    IF money_request.amount > cert.certified_amount:
      BLOCK "Exceeds certified amount (SAR X)."
      System prevents overpayment above certified amount
```

G4 — Invoice Submission on PO Exhaustion

Purpose: Forces formal invoicing cycle when PO is fully paid. Ensures ZATCA compliance. Accounts can reconcile PO vs invoice vs payments.

Override: DH can attach invoice on SubCon behalf if SubCon is unable.

DECISION LOGIC

```
WHEN:     SubCon tries next payment request OR DH submits MR
CHECK:
  total_paid = SUM(money_request_lines.amount WHERE po_id=X AND status="PAID")
  po_value   = outgoing_purchase_orders.value

  IF total_paid >= po_value:
    invoice_present = outgoing_purchase_orders.invoice_url IS NOT NULL

    IF NOT invoice_present:
      BLOCK ALL further payment requests for this PO
      NOTIFY SubCon: In-App + Telegram
        "PO fully paid. Submit invoice to proceed."
      NOTIFY DH: "PO #X exhausted. Invoice required."
      ALLOW: DH can attach invoice on SubCon behalf

  AFTER INVOICE ATTACHED:
    UPDATE outgoing_purchase_orders SET
      invoice_url=?, invoice_date=?, status="INVOICED"
    NOTIFY Accounts: Invoice → ZATCA validation queue
    New PO can now be issued if further work needed
```

G5 — QR Authentication (All Field Forms)

Purpose: All field forms require valid QR token. No QR = no form access. Tokens managed via QR Manager (desktop). All access is logged.

Override: Desktop login (DH/PM) bypasses QR for desktop forms. QR is for field/mobile only.

DECISION LOGIC

```
WHEN:     Any field form URL opened (mobile/tablet)
CHECK:
  token  = URL param OR QR scan result
  record = SELECT * FROM qr_tokens WHERE token=?

  IF record = NULL    → BLOCK "Invalid QR code."
  IF record.expiry < NOW() → BLOCK "QR code expired."
  IF record.active = FALSE → BLOCK "QR code deactivated."

RESOLVE ACTOR:
  IF record.actor_type = "EMPLOYEE"   → load employee, filter forms by role
  IF record.actor_type = "CONTRACTOR" → load contractor, SubCon forms only
```

```
  IF record.actor_type = "DH"          → load DH, show DH forms

PREFILL: entity_id, site_id, actor_id on all forms (no manual entry)
AUDIT:   log token use → qr_access_log {token_id, form, timestamp, ip}
```

G6 — Payment Validation (Employee Confirms Cash Advance)

Purpose: Creates paper trail for cash distribution. Reduces payment disputes. Signed voucher photo = physical evidence of receipt.

Override: DH can mark confirmed manually with audit note (e.g. paper signed but employee has no phone).

DECISION LOGIC

```
WHEN:     Cash advance issued via FormFieldPayment

IMMEDIATE:
  NOTIFY Employee: In-App "SAR X advance issued. Confirm receipt."
  employee_advances.confirmed = FALSE ← flagged on DH dashboard

24h MONITORING:
  IF confirmed = FALSE after 24 hours:
    ALERT DH: "[Employee] has not confirmed SAR X advance"
    Flag persists until confirmed

WHEN EMPLOYEE CONFIRMS:
  UPDATE employee_advances:
    confirmed=TRUE, confirmation_photo=URL, confirmed_at=NOW()
  DH dashboard flag removed
```

G7 — OT Approval Gate (Before Payroll)

Purpose: Forces DH to consciously approve or reject each OT claim. Payroll cannot close with pending unapproved OT records.

Override: None. All OT must be explicitly approved by DH.

DECISION LOGIC

```
WHEN:     Monthly payroll run initiated
CHECK:
  pending_ot = SELECT * FROM overtime_records
               WHERE status="PENDING" AND month=payroll_month

  IF pending_ot EXISTS:
    EXCLUDE from payroll calculation
    DH DASHBOARD: shows list of pending OT for action

  DH ACTIONS per record:
    Approve → status="APPROVED" → included in payroll
    Reject  → status="REJECTED" → excluded, employee notified

  PAYROLL FORMULA:
    ot_pay = hours × hourly_rate × multiplier
    Multiplier: NORMAL/FRIDAY/HOLIDAY=1.5, TRAVEL=1.0

  SAFETY: Payroll cannot close month with pending (unapproved) OT.
```

G8 — Expense Claim DH Approval Gate

Purpose: Accounts cannot see or pay any expense claim until DH approves. Ensures DH validates receipts and amounts before Accounts processes.

Override: DH can partially approve (edit amounts). Rejected claims return to DRAFT for employee correction.

DECISION LOGIC

```
WHEN:     Employee submits monthly expense claim
STATUS:   expense_claims.status = "SUBMITTED"

STEP 1:  NOTIFY DH: Telegram + In-App

STEP 2:  DH reviews each line: receipt photos, amounts, categories

DH ACTION:
```

```
  Approve → status="APPROVED"
    → post_je(): Dr Expense Account / Cr Employee Payable
    → GENERATE Expense Claim DOCX ▫
    → NOTIFY Employee: Email + In-App "Approved"
    → Claim VISIBLE to Accounts for payment

  Reject  → status="REJECTED"
    → NOTIFY Employee with rejection reason
    → Employee corrects → resubmits

  Partial → DH edits amounts → partial approval

ACCOUNTS RULE:
  Accounts CANNOT see claim until DH approval.
  Payment queue shows ONLY approved claims.

CUTOFF:
  Submission closed after monthly_cutoff_date.
```

SECTION 3 — Notification Matrix

Every system event with its notification channel(s), recipient(s), and message content.

Event	Telegram	Email	In-App Alert	Template
PO issued to SubCon	✓ SubCon	✓ SubCon (□ template missing)	✓ SubCon: "New PO #X. Tap to acknowledge."	Outgoing PO DOCX □ attached
SubCon acknowledges PO	—	—	✓ DH: "[SubCon] acknowledged PO #X"	None
Incoming PO from Client	—	✓ Accounts (□ template missing)	✓ Accounts: "Client PO received"	Incoming PO Template □
SubCon payment request submitted	✓ DH	—	✓ DH: "Payment request from [SubCon] — SAR X"	None
Payment Certificate created	—	—	✓ Accounts: "MR can now be submitted"	SubCon Payment Certificate ⚠
Money Request submitted	✓ Accounts	✓ Accounts (□ template missing)	✓ Accounts: "New MR #X — SAR Y"	Money Request Template □
Payment made to SubCon	✓ SubCon	✓ DH (□ template missing)	✓ SubCon: "SAR X credited. PO balance: SAR Y"	Email: Payment Made □
PO exhausted — invoice required	✓ SubCon + DH	—	✓ SubCon: "Submit invoice." ✓ DH: "Invoice required."	None — invoice upload triggered
Invoice received from SubCon	—	✓ Accounts	✓ Accounts: "Invoice received for PO #X"	ZATCA validation
Site Assignment created	—	✓ Employee (□ template missing)	✓ Employee: "You are assigned to Site [X]"	Email: Site Assignment □
Cash advance issued (Field Payment)	—	—	✓ Employee: "SAR X advance issued. Confirm receipt."	None — employee confirms via form
Payment validation received	—	—	✓ DH: "[Employee] confirmed SAR X advance"	Signed voucher photo in vault
Advance NOT	—	—	✓ DH: "⚠ [Employee]	Dashboard flag —

Event	Telegram	Email	In-App Alert	Template
confirmed (24h)			has not confirmed SAR X"	persistent
OT submitted	—	—	✓ DH: "OT pending: [Employee] [Xh] [Type]"	None
OT approved	—	—	✓ Employee: "OT approved — SAR X"	None
Food Allowance submitted	—	—	✓ DH: "Food allowance pending: SAR X"	None
Food Allowance approved	—	—	✓ Employee: "Food allowance approved"	None
Expense claim submitted	✓ DH	—	✓ DH: "Expense claim from [Employee] — SAR X"	None (pending DH approval)
Expense claim approved	—	✓ Employee (□ template missing)	✓ Employee: "Expense claim approved — SAR X"	Expense Claim DOCX □
Payment made to Employee	—	✓ DH (□ template missing)	✓ Employee: "SAR X payment processed"	Email: Payment Made □
Invoice sent to client	—	✓ Client (□ template missing)	✓ Accounts: "Invoice #X sent to [Client]"	ZATCA Invoice Template □
Invoice paid by client	—	—	✓ Accounts: "Payment received for Invoice #X"	Receipt issued

SECTION 4 — Table Dependency Map

What each form reads and writes. Shared tables create inter-form dependencies.

Form	Writes To	Reads From	Shared Table / Dependency
FormSubConClaim	subcon_claims	outgoing_purchase_orders, sites, contractors, qr_tokens	outgoing_purchase_orders — also read by FormMoneyRequest, Payment Certificate
Payment Certificate	payment_certificates	subcon_claims, outgoing_purchase_orders, site_completion_records	payment_certificates — read by FormMoneyRequest (G3 check)
FormMoneyRequest	money_requests, money_request_lines	employees, contractors, suppliers, sites, projects, outgoing_purchase_orders, payment_certificates, subcon_claims, bank_accounts	money_request_lines — reduces outgoing_purchase_orders. balance after payment
FormFieldPayment	employee_advances (ADVANCE_IN), journal_entries	employees, bank_accounts, sites	employee_advances — updated by FormPaymentValidation
FormPaymentValidation	employee_advances (UPDATE confirmed=TRUE)	employee_advances (unconfirmed only)	Must exist from FormFieldPayment

Form	Writes To	Reads From	Shared Table / Dependency
FormFoodAllowance	food_allowances	employees, sites, departments	food_allowances — compiled into FormExpenseClaim
FormOvertime	overtime_records	employees, sites, departments	overtime_records — read by Payroll run after DH approval
FormDailyExpense	daily_expenses	employees, sites, expense_categories	daily_expenses — compiled into FormExpenseClaim at month-end
FormExpenseClaim	expense_claims, expense_claim_lines, journal_entries	daily_expenses, employees, sites, food_allowances, employee_advances	expense_claims — read by Accounts payment queue (after DH approval)
FormCarMaintenance	vehicle_maintenance	vehicles, employees, sites	vehicle_maintenance — independent, feeds Vehicle Report
FormSiteCompletion	site_completion_records, milestone_records	sites, milestones, projects, subcon_assignments	site_completion_records — read by Payment Certificate (certification basis)
FormPORequest	po_requests → outgoing_purchase_orders	departments, employees, projects, sites, contractors, suppliers, items	outgoing_purchase_orders — central shared table (8 forms/processes touch it)
FormPayment (Shareable)	payments (travel / field distribution)	employees, contractors, bank_accounts, qr_tokens	payments — feeds ledger accounts by payment_type

SECTION 5 — Inter-Relation Map

Chain 1: SubCon Payment Cycle

```
FormPORequest
  └──► outgoing_purchase_orders (OPEN, balance = po_value)
          ├──► [G1] SubCon acknowledges PO
          └──► FormSubConClaim (SubCon requests payment)
                  ├──► FormSiteCompletion (DH checks site progress %)
                  └──► Payment Certificate (DH certifies work % + amount)
                          └──► [G3 UNLOCKED] FormMoneyRequest (SubCon line)
                                  └──► Accounts pays → UPDATE po.balance
                                          ├──► [G4] IF po.balance=0 → Invoice gate
                                          │         SubCon submits invoice → ZATCA
                                          └──► SubCon In-App credit notification
```

Chain 2: Employee Expense Cycle

```
Site Assignment (Desktop)
  └──► Employee assigned to site (email notification)
          ├──► FormFoodAllowance (daily, per site)
          │       └──► DH approves [G7 subset] → Payroll
          ├──► FormOvertime (daily/weekly)
          │       └──► [G7] DH approves → Payroll
          └──► FormDailyExpense (daily field costs)
```

```
          └──► Accumulates → month-end

    └──► FormExpenseClaim (month-end)
           Compiles: daily_expenses + food_allowances (approved)
           [G8] DH approves → Auto JE + Expense Claim DOCX ▫
                  └──► Accounts payment queue
```

Chain 3: Cash Advance Cycle

```
FormFieldPayment (DH issues advance)
  └──► employee_advances (ADVANCE_IN)
       Auto JE: Dr 1310 Employee Advance / Cr Petty Cash
         │
         └──► [G6] FormPaymentValidation (employee confirms)
                └──► confirmed=TRUE, photo stored in vault
                       │
                       └──► FormExpenseClaim (month-end advance reconcile)
```

Critical Shared Table: outgoing_purchase_orders

This single table is the central hub of the SubCon payment cycle. 8 different forms and processes read or write it:

Form / Process	Relationship	Field / Purpose
FormPORequest	CREATES record	All fields. status="OPEN", balance=po_value
SubCon Acknowledgement	UPDATES	acknowledged=TRUE, acknowledged_at=NOW()
FormSubConClaim	READS (G2 check)	status, balance, contractor_id
Payment Certificate	READS	value, prev_paid → calculates certifiable amount
FormMoneyRequest	READS (G3 check)	po_id for line linking
After Accounts Pays	UPDATES	balance = balance − paid_amount
G4 (PO Exhaustion)	READS	balance=0 → triggers invoice gate
Invoice Submission	UPDATES	invoice_url, invoice_date, status="INVOICED"
BankRecon (month-end)	READS	Cross-checks actual bank debit vs po payments

SECTION 6 — Bugs, Patches & Missing Items

Known Code Bugs (Patch Session)

BUG 1 — FormMoneyRequest: project_id not saved to lines File: src/forms/FormMoneyRequest.jsx EMPTY_LINE has project_id in state but lineRows insert omits it. Fix: add project_id: l.project_id || null, site_id: l.site_id || null to lineRows map.

BUG 2 — FormMoneyRequest: No Site# field on money_request_lines Without Site#, system cannot auto-resolve which open PO to apply SubCon payment against. Fix: Add site_id selector per line (cascade Project → Site → open POs for SubCon).

Code Patches (Planned)

PATCH 1 — Simplify FormSubConClaim to 3 taps: site picker + amount + photo. Remove claim_type, milestones, hours, hourly_rate, scope_type, work_description. DH fills complex fields in Payment Certificate, not SubCon.

PATCH 2 — Extend FormFieldPayment to support SubCon as payee type + link to open Outgoing PO.

PATCH 3 — Excel UserForm: one Excel file, UserForms pulling Project/Site/SubCon from Supabase. VBA submits to money_requests table (same as App). Interface only — same backend.

PATCH 4 — Enforce G1/G2/G3/G4 at code level in FormSubConClaim + FormMoneyRequest submission handlers.

Missing Templates (Build Queue)

Template Name	Status	Description / What Needs Building
SubCon Payment Certificate ⚠	**REBUILD NEEDED**	Add work_progress_pct, PO status block (Total / Previously Paid / This Payment / Balance After), earned vs certified calculation.
Incoming PO Template □	**MISSING**	Client PO acknowledgement and receipt confirmation. Triggers AR cycle.
Money Request Template □	**MISSING**	Printable MR document for Accounts physical records and approval workflow.
Standard Tax Invoice (ZATCA) □	**MISSING**	B2B client invoice. ZATCA compliant. QR code required.
Simplified Tax Invoice (ZATCA) □	**MISSING**	B2C / small transactions. ZATCA simplified format.
Credit Note □	**MISSING**	Adjustment to issued invoice. ZATCA required.
Debit Note □	**MISSING**	Upward adjustment to issued invoice. ZATCA required.
Proforma Invoice □	**MISSING**	Pre-billing estimate for client approval before formal invoice.
Email: Site Assignment □	**MISSING**	Notify employee/SubCon of site assignment with dates and site details.
Email: PO Issued to SubCon □	**MISSING**	Formal PO issuance email with Outgoing PO PDF attachment.
Email: MR Submitted □	**MISSING**	Notify Accounts of new Money Request from DH with summary.
Email: Payment Made □	**MISSING**	Confirm payment to DH + SubCon/Employee. Amount, reference, date.
Email: Expense Claim Approved □	**MISSING**	DH approval confirmation to employee with claim summary.
Email: Invoice Sent to Client □	**MISSING**	Cover email for client invoice submission with ZATCA invoice attached.

Missing Relationships (To Add)

Bank Reconciliation ↔ outgoing_purchase_orders: BankRecon should flag unmatched bank debits against open PO payments. ATM-bypass payments by DH appear as unmatched → red flag to Finance.

FormSiteCompletion → FormExpenseClaim: Site completion % could auto-validate whether field expenses are reasonable for the work volume claimed on site.

FormPORequest → Incoming PO: System needs a mechanism to record client POs (Incoming). Currently only Outgoing POs are managed. Incoming PO is the AR trigger for client billing.

FormOvertime → FormSiteCompletion: OT hours on a site should align with site activity log. Cross-validation possible to flag OT submitted on inactive sites.